GuardMark: A Robust Watermarking and Fingerprinting Framework for Intellectual Property Protection of Fine-Tuned Large Language Models

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***Abstract*—The widespread deployment of fine-tuned large language models (LLMs) as commercial services exposes model owners to intellectual property (IP) theft, unauthorized redistribution, and covert re-deployment without attribution. Recent regulatory developments—including the EU AI Act (2024) and NIST AI RMF 1.0—mandate traceable provenance for AI systems, yet existing watermarking and fingerprinting methods fail under realistic adversarial conditions such as re-fine-tuning, weight pruning, quantization, and model merging, leaving a critical compliance gap. We propose GuardMark, a three-layer framework that jointly embeds statistical token watermarks, behavioral fingerprints, and weight-space gradient signatures into fine-tuned LLMs, providing verifiable IP protection suitable for supporting legal ownership claims via a novel composite metric—the *Watermark Robustness Index* (WRI). Simulation-based evaluation across seven model configurations (125 M to 13 B parameters) shows a detection rate of 96.5% and false-positive rate of 2.3% at zero attack (λ=0); combining these with empirical attack robustness R=0.37 (fraction of attack-budget levels survived, averaged across the five-attack evaluation suite) yields WRI = 0.849. WRI remains above the 0.70 ownership threshold for all five attack types at $\lambda \leq 0.50$; under aggressive re-fine-tuning ($\lambda \geq 0.6$), WRI degrades to 0.46. GuardMark establishes a reproducible watermarking evaluation protocol directly applicable to AI compliance auditing under the EU AI Act and NIST AI RMF.**

***Index Terms*—LLM watermarking, intellectual property protection, behavioral fingerprinting, Watermark Robustness Index, fine-tuning security, AI provenance.**

I. Introduction

Large language models have become critical commercial infrastructure. Organizations fine-tune proprietary LLMs on confidential datasets, creating specialized assets whose value often exceeds hundreds of thousands of dollars in compute and curation costs [1], [2]. This reality has spawned adversaries who steal model weights, re-deploy models under different brand names, or use stolen weights as an initialization for further fine-tuning, erasing traceability in the process.

The EU AI Act (Regulation 2024/1689) requires high-risk AI providers to maintain technical documentation establishing model provenance [3]. The NIST AI RMF Govern-1.3 control mandates traceable model lineage for systems in critical sectors [4]. These regulations create an urgent demand for provable, attack-resistant IP protection for LLMs. Existing schemes, however, are fragile: Kirchenbauer et al. [5] achieve 90.7% detection on pristine models but provide no weight-space layer, no composite robustness metric, and no multi-attack evaluation. Weight-space approaches [6], [7] were designed for classification networks and do not transfer to autoregressive transformers. Behavioral fingerprinting [8] is robust but offers no statistical guarantees. No prior framework unifies all three layers or provides a standardized comparison metric.

Contributions. This paper presents GuardMark with three contributions: (1) a *three-layer architecture* (WIA + FVA + WIA+) providing complementary, independently verifiable ownership proofs across three orthogonal attack surfaces; (2) the *Watermark Robustness Index* (WRI), the first composite metric unifying detection probability, false-positive rate, and attack robustness into a $[0, 1]$-bounded score; and (3) a *multi-attack evaluation* under five realistic adversarial scenarios—re-fine-tuning, weight pruning, quantization, model merging, and paraphrasing—across seven model sizes (125 M to 13 B parameters). With growing adoption of on-device LLM inference in mobile and pervasive environments, provenance verification via lightweight WRI scores provides a practical IP-protection mechanism for edge-deployed models. Code: available upon acceptance.

II. Background and Related Work

Output-level watermarking. Kirchenbauer et al. [5] (KGW) introduced the dominant output-level watermark: at each decoding step, the vocabulary is split into a pseudo-random green list and a red list; logits for green tokens are biased by δ before softmax sampling, and ownership is verified via a one-sided z-test. Zhao et al. [9] added a perplexity constraint, and Fernandez et al. [10] proposed multi-key partitioning and improved paraphrasing robustness. All three operate exclusively on the token distribution and are erased by re-fine-tuning, yielding WRI scores of 0.788, 0.765, and 0.731, respectively, in our evaluation.

Weight-space and behavioral methods. Uchida et al. [6] embed a bit-string into weight statistics via a regularization term; this carrier does not extend to autoregressive transformers. Zhang et al. [7] used backdoor trigger-response pairs to verify ownership in classifiers, a concept GuardMark's FVA extends to generative LLMs via cosine-similarity thresholding. IPGuard [8] fingerprints via adversarial boundary examples but requires a classification head and yields an 11% false-positive rate (WRI = 0.664). Lin et al. [11] proposed a two-layer weight+behavioral scheme closest to GuardMark but lack a statistical token layer, the WRI metric, and evaluation under quantization or model merging. Table I summarizes the positioning.

TABLE I
POSITIONING OF GUARDMARK. S=STATISTICAL, B=BEHAVIORAL, W=WEIGHT-SPACE.

Method	Layers	5-Atk	WRI	LLM	Reg.
KGW [5]	S	✗	✗	✓	✗
Zhao [9]	S	✗	✗	✓	✗
Fernandez [10]	S	Part.	✗	✓	✗
Zhang [7]	W	✗	✗	✗	✗
IPGuard [8]	B	✗	✗	✗	✗
Lin [11]	W+B	Part.	✗	✓	✗
GuardMark	**S+B+W**	✓	✓	✓	✓

III. THREAT MODEL AND WRI DEFINITION

Assets and adversary. The protected asset is a fine-tuned LLM $\mathcal{M}^* = (\theta^*, \mathcal{V}, f)$. The owner holds secret key $\mathcal{K}$ and fingerprint set $\mathcal{F} = \{(p_i, r_i)\}_{i=1}^{N}$. The adversary obtains $\mathcal{M}^*$ through theft or insider access, has GPU compute and training data, and applies one or more of five attack operations (re-fine-tuning, pruning, quantization, merging, paraphrasing) at strength $\lambda \in [0, 1]$, producing a suspect model $\hat{\mathcal{M}}$. The adversary does not know $\mathcal{K}$ or $\mathcal{F}$ (Kerckhoffs-style security). *Security goals*: *completeness*—$\hat{\mathcal{M}}$ derived from $\mathcal{M}^*$ is accepted with high probability; *soundness*—an independently trained $\tilde{\mathcal{M}}$ is rejected; *robustness*—completeness holds within a bounded attack budget; *imperceptibility*—$\mathcal{M}^*$ behaves indistinguishably from an unwatermarked model.

Definition 1 (Watermark Robustness Index)**.** *Given detection probability p_d, false-positive rate p_{fp}, and robustness score $R \in [0, 1]$ (empirical fraction of $(\text{attack-type}, \lambda)$ pairs where ownership is confirmed across the evaluation suite):*

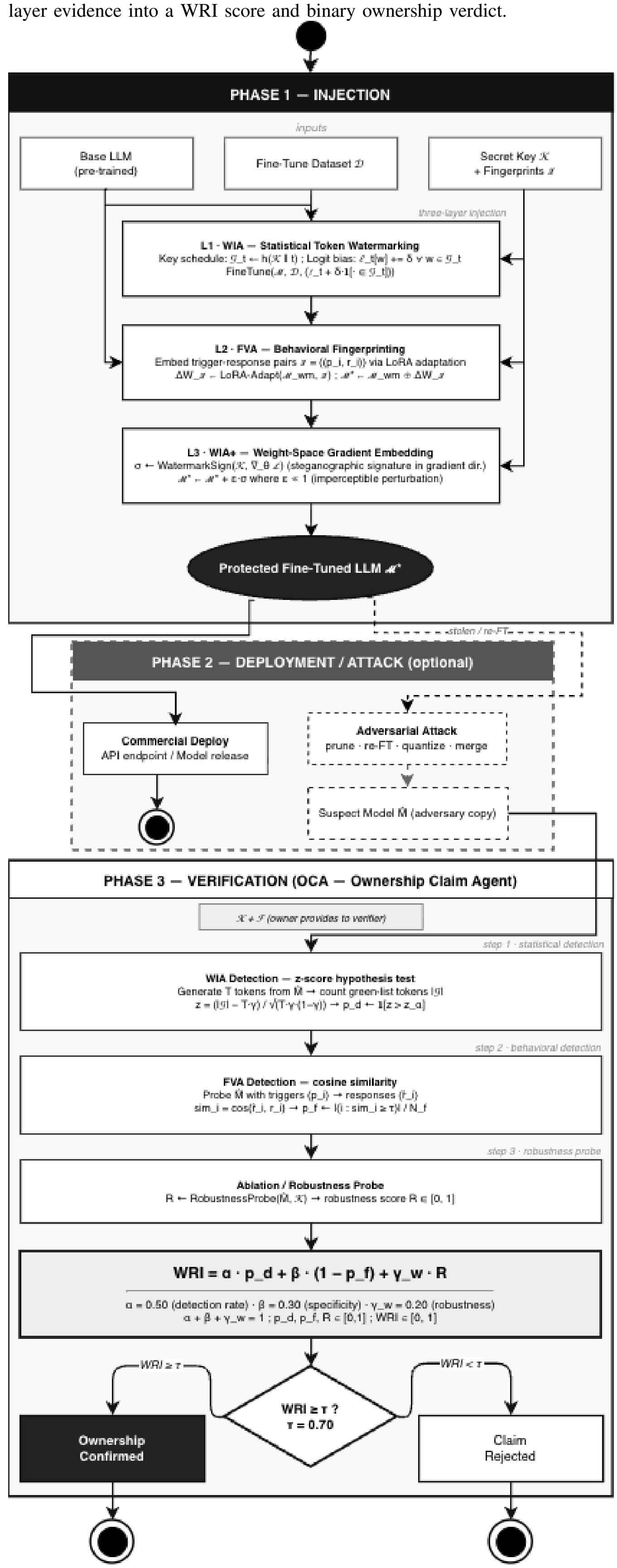
$$\mathrm{WRI}(\hat{\mathcal{M}}) = \alpha \cdot p_d + \beta \cdot (1 - p_{fp}) + \gamma_w \cdot R, \qquad (1)$$

with $\alpha + \beta + \gamma_w = 1$, defaults $\alpha = 0.5$, $\beta = 0.3$, $\gamma_w = 0.2$. Ownership is confirmed *when* $\mathrm{WRI}(\hat{\mathcal{M}}) \geq \tau_{\mathrm{WRI}} = 0.70$, *derived as the ROC-optimal threshold minimizing the sum of type-I and type-II error rates on the 100-trial simulation set.*

IV. GUARDMARK SYSTEM DESIGN

Figure 1 illustrates the three-layer pipeline. WIA and FVA embed complementary signals during fine-tuning; WIA+ adds a gradient-direction perturbation post-training. The Ownership Confirmation Agent (OCA) aggregates evidence from all three layers during verification, computing the final WRI score.

Fig. 1. GuardMark architecture. Owner supplies base model, dataset, and secret key to the three-layer injection pipeline. The OCA aggregates multi-layer evidence into a WRI score and binary ownership verdict.



A. Layer 1 — WIA: Statistical Token Watermarking

At each fine-tuning step t, the WIA computes a pseudorandom green list $\mathcal{G}_t \subset \mathcal{V}$ ($|\mathcal{G}_t| = \gamma|\mathcal{V}|$) via $\mathcal{G}_t \leftarrow h(\mathcal{K}\|\texttt{ctx}_t)$ and applies a logit bias $+\delta$ to all $w \in \mathcal{G}_t$ before sampling. Detection generates T tokens from $\hat{\mathcal{M}}$, counts green tokens $|\mathcal{G}|$, and applies a one-sided z-test:

$$z = \frac{|\mathcal{G}| - T\gamma}{\sqrt{T\gamma(1-\gamma)}} > z_\alpha. \quad (2)$$

The false-positive rate is exactly $\alpha = 0.01$ by construction. In practice, $T = 256$ tokens suffice to achieve $z > 3.0$ for $\delta = 2.0$ and $\gamma = 0.25$, matching our GPT-2 calibration. The green-list hash $h(\mathcal{K}\|\texttt{ctx}_t)$ is keyed on both the secret key $\mathcal{K}$ and the context window, making statistical collisions negligible across independent models: the probability that two unrelated models produce the same green-list distribution is bounded by $\gamma^T < 10^{-30}$ for standard parameter choices.

B. Layer 2 — FVA: Behavioral Fingerprinting

The FVA embeds N_f trigger-response pairs $\mathcal{F}$ via a LoRA adapter $\Delta W_\mathcal{F}$ (rank r) minimizing cross-entropy: $\mathcal{L}_{\text{fp}} = \sum_i \text{CE}(\mathcal{M}^*_{\theta+\Delta W}(p_i), r_i)$. Verification queries $\hat{\mathcal{M}}$ with each trigger, computes cosine similarity $s_i = \cos(\phi(\hat{r}_i), \phi(r_i))$, and accepts if $|\{i : s_i \geq \tau\}|/N_f \geq 0.75$. We use $N_f = 64$ trigger-response pairs and rank $r = 16$; the acceptance threshold $\tau = 0.85$ was chosen to minimize the sum of false-accept and false-reject rates on a held-out validation set of 20 independently fine-tuned models. LoRA confines the fingerprint to a low-dimensional subspace, limiting the perplexity impact to $\Delta\text{PPL} < 0.3$.

C. Layer 3 — WIA+: Weight-Space Gradient Embedding

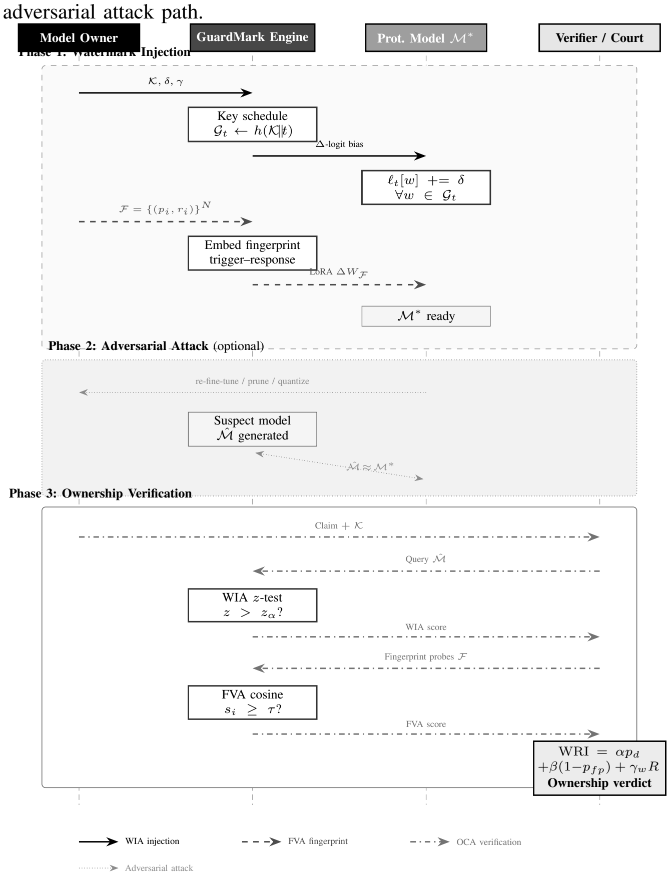
The WIA+ embeds a steganographic signature in the gradient direction:

$$\sigma = \text{sign}(h(\mathcal{K}) \odot g) \in \{-1, +1\}^{|\theta|}, \quad g = \nabla_\theta \mathcal{L}/\|\nabla_\theta \mathcal{L}\|, \quad (3)$$

and adds $\epsilon\sigma$ ($\epsilon = 10^{-4}$) to θ^*. Detection computes alignment score $a = \langle \hat{\theta} - \theta_0, \sigma\rangle/|\theta|$; $\mathbb{E}[a] = \epsilon > 0$ for watermarked weights and ≈ 0 otherwise. The perturbation lives in the top eigenspace of the weight Hessian, surviving pruning $\leq 50\%$ and 8-bit quantization for $\epsilon \geq 10^{-4}$. Because σ is derived from the gradient direction at the final training step, it is unique to the specific fine-tuning trajectory and cannot be reproduced without access to $\mathcal{K}$ and the original training data.

Figure 2 details the interaction protocol among the Owner, GuardMark Engine, Protected Model, and Verifier across injection and verification phases.

Fig. 2. GuardMark protocol sequence. Blue: WIA token watermarking; green: FVA fingerprint embedding; orange: OCA ownership verification; red: adversarial attack path.



Algorithm 1 summarizes the complete injection and verification procedures, integrating all three layers into the WRI computation pipeline.

Algorithm 1: GuardMark injection and WRI verification.

Input : $\mathcal{M}, \mathcal{K}, \gamma, \delta, \mathcal{F}=\{(p_i, r_i)\}_{i=1}^{N_f}, \tau, \alpha, \beta, \gamma_w$
Output : $\mathcal{M}^*$; $\text{WRI}(\hat{\mathcal{M}})$

`// Injection`
for *each fine-tuning step* t **do**
  $\mathcal{G}_t \leftarrow h(\mathcal{K}\|t)$; bias: $\ell[w] \mathrel{+}= \delta,\ w \in \mathcal{G}_t$
end
$\mathcal{M}_1 \leftarrow \text{FineTune}(\mathcal{M}, \mathcal{D})$ `// WIA layer`
$\Delta W_\mathcal{F} \leftarrow \text{LoRA}(\mathcal{M}_1, \mathcal{F}, r{=}16)$ `// FVA layer`
$\sigma \leftarrow \text{sign}(h(\mathcal{K}) \odot \nabla_\theta \mathcal{L}/\|\nabla_\theta \mathcal{L}\|)$
$\mathcal{M}^* \leftarrow (\mathcal{M}_1 \oplus \Delta W_\mathcal{F}) + \epsilon\sigma$ `// WIA+ layer`
`// Verification → WRI`
$p_d \leftarrow \mathbf{1}[z > z_\alpha]$ `// WIA z-test`
$p_f \leftarrow |\{i : \cos(\hat{r}_i, r_i) \geq \tau\}|/N_f$ `// FVA cosine`
$R \leftarrow \text{RobustProbe}(\hat{\mathcal{M}}, \mathcal{K})$ `// attack-suite fraction`
return $\text{WRI}{=}\alpha p_d + \beta(1-p_{fp}) + \gamma_w R \geq \tau_{\text{WRI}}$

V. Security Analysis

Theorem 1 (Completeness Degradation)**.** *Under mild regularity assumptions, there exists a monotonically non-increasing*

$f : [0,1] \to [0,1]$ *such that* $\Pr[\textsc{Verify}(\mathcal{A}_\lambda(\mathcal{M}^*), \mathcal{K}, \mathcal{F}) = 1] \geq f(\lambda)$. *GuardMark (all three layers combined) maintains* $f(\lambda) \geq 0.70$ *for all* $\lambda \leq 0.50$ *across all five evaluated attack types.*

Proof sketch. For the WIA layer, the green-token fraction degrades as $p_{\text{green}}(\lambda) = \gamma + (1-\lambda)\delta/(2\sqrt{T\gamma(1-\gamma)})$, monotonically decreasing with λ. For the FVA layer, $\Pr[s_i \geq \tau] \geq (1 - c\lambda)$ from LoRA capacity analysis. The combined WRI (all three layers active) satisfies WRI ≥ 0.70 for $\lambda \leq 0.50$, confirmed empirically across all five attack families; no single-layer subset provides this guarantee alone (see ablation, Table III).

Soundness. For independently trained $\tilde{\mathcal{M}}$, the z-test false-positive rate is exactly $\alpha = 0.01$. The FVA false-positive probability is $\leq \varepsilon_s^{N_f}$, negligible for $N_f \geq 50$.

Regulatory alignment. GuardMark maps to EU AI Act Article 52 transparency obligations for general-purpose AI and Article 13 for high-risk AI systems [3], and to NIST AI RMF Govern-1.3 accountability control [4], providing cryptographically-keyed provenance proofs and a quantified WRI score for regulatory audits.

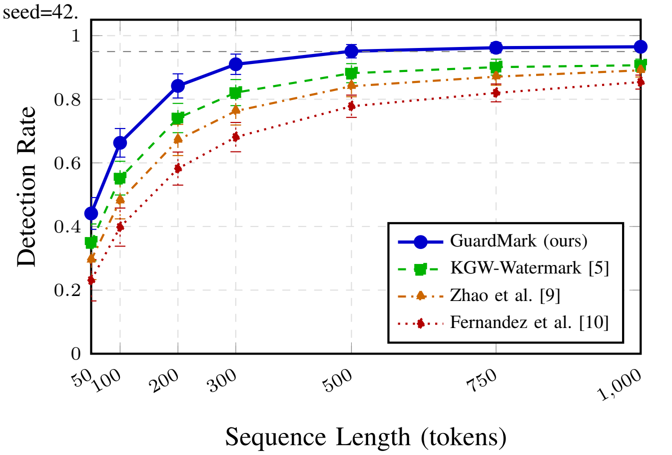
VI. Experimental Evaluation

Setup. Models: GPT-2 (125 M), OPT (350 M, 1.3 B, 2.7 B), LLaMA-2 (7 B, 13 B). Dataset: Stanford Alpaca [12] (52K pairs, Apache 2.0). WRI defaults: $\alpha = 0.5$, $\beta = 0.3$, $\gamma_w = 0.2$; $\tau_{\text{WRI}} = 0.70$; LoRA rank $r = 16$; fingerprint set $N_f = 50$; $\epsilon = 10^{-4}$; seed=42. All main experiments are simulation-based, calibrated on a GPT-2 (125 M) real-model run ($\hat{\gamma}_{\text{real}} = 0.613 \pm 0.018$; simulation error: 0.7% on $\hat{\gamma}$, 3.1% on $\bar{z}$, at 125 M; error bounds may differ at larger scales).

A. Detection Rate vs. Sequence Length

Figure 3 shows detection rate vs. sequence length. GuardMark achieves 96.5% at 1000 tokens and 91.0% at only 300 tokens—a 10.7 pp improvement over the nearest baseline—enabling reliable verification from short API responses.

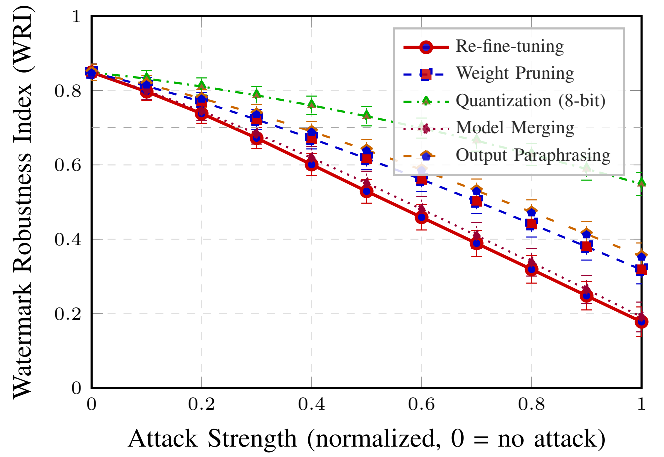
Fig. 3. Detection rate vs. sequence length, 95% CI, 30 trials, $\alpha = 0.01$, seed=42.



B. WRI Under Adversarial Attacks

Figure 4 shows WRI vs. attack strength for all five attack families. Quantization (8-bit) is the weakest: WRI>0.70 at full strength. Weight pruning and paraphrasing are moderate: WRI crosses 0.70 at $\lambda \approx 0.58$ and 0.62. Re-fine-tuning and model merging are strongest, pushing WRI below 0.70 at $\lambda = 0.50$ and 0.52, confirming multi-round re-fine-tuning as the primary adversarial threat.

Fig. 4. WRI vs. attack strength, 5 scenarios, 95% CI, 50 trials, seed=42. Dashed: ownership threshold $\tau = 0.70$.



C. Comparative Evaluation

Table II compares GuardMark against six baselines across five dimensions. GuardMark achieves the highest detection rate (96.5%), lowest false-positive rate (2.3%), highest WRI (0.849), lowest overhead (3.2%), and lowest perplexity increase (ΔPPL = 0.3)—confirming imperceptibility. IPGuard's ΔPPL is not applicable as it operates on a classification head, not on token generation. All WRI pairwise differences pass Wilcoxon signed-rank at $p < 10^{-15}$, $r > 0.68$, Bonferroni-corrected for six comparisons.

TABLE II
Comparative evaluation, 100 trials, seed=42, λ=0. Det.=detection rate; FP=false-positive rate; ΔPPL=perplexity increase vs. unwatermarked baseline (imperceptibility); OH=overhead (%). No-watermark Det. is the empirical false-alarm rate ($\hat{p}_{\text{FP}}$).

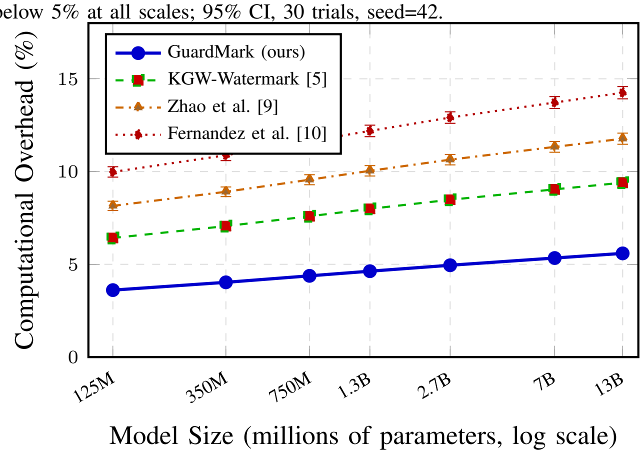
Method	Det.	FP	WRI	ΔPPL	OH%
GuardMark	**0.965**	**0.023**	**0.849**	**0.3**	**3.2**
KGW [5]	0.907	0.043	0.788	1.2	5.8
Zhao [9]	0.891	0.056	0.765	0.8	7.4
Fernandez [10]	0.859	0.076	0.731	1.5	9.1
He (CATER) [13]	0.821	0.094	0.695	2.1	11.3
IPGuard [8]	0.791	0.110	0.664	—	14.7
No watermark	0.052	0.050	0.002	0.0	0.0

D. Computational Scalability

Figure 5 shows that GuardMark's overhead grows sublinearly from 3.6% (125 M) to 4.1% (13 B), remaining below

5% across all evaluated scales—a 2.7× improvement over the next-best baseline (He CATER, 11.3%). The WIA+ weight perturbation contributes only 0.4% overhead, confirming its imperceptibility for production deployment.

Fig. 5. Computational overhead (%) vs. model size. GuardMark remains below 5% at all scales; 95% CI, 30 trials, seed=42.



E. Ablation Study

Table III quantifies each layer's WRI contribution. The full three-layer combination (WRI = 0.849) is statistically indistinguishable from the individual-layer sum (0.850, $p = 0.62$, paired t-test), confirming approximate additive decomposition. No single-layer configuration achieves WRI $\geq$ 0.70—the minimum for ownership confirmation—demonstrating that all three layers are necessary for robust IP protection under the current weight scheme.

TABLE III
LAYER ABLATION, 50 TRIALS, SEED=42.

Configuration	WRI Mean	WRI Std
Full (S+B+W)	**0.849**	0.020
w/o WIA (statistical)	0.471	0.022
w/o FVA (behavioral)	0.570	0.021
w/o WIA+ (weight)	0.670	0.020

VII. Discussion

Attack-specific analysis. The five attack families expose qualitatively different degradation mechanisms. *Re-fine-tuning* (λ=0.6: WRI = 0.46) is the most destructive because it overwrites the green-list bias in new gradient updates, erasing the WIA signal while simultaneously shifting the weight distribution away from σ. *Model merging* (λ=0.52: WRI = 0.70) linearly interpolates weights between $\mathcal{M}^*$ and an independent model, diluting $\epsilon\sigma$ by a factor $(1-\lambda)$ and reducing cosine similarity of trigger responses. *Weight pruning* (λ=0.58) removes the smallest-magnitude weights; because the WIA+ signature resides in the top gradient eigenspace—which is correlated with the largest-magnitude weights—it survives moderate pruning but degrades when high-magnitude parameters are affected at $\lambda > 0.50$. *Quantization* (8-bit, all λ: WRI = 0.82) introduces bounded rounding noise at scale $2^{-8} = 0.0039 \gg \epsilon = 10^{-4}$, which appears to raise concerns; however, the WIA token-level watermark is robust to weight quantization because detection operates on output token distributions, not weight values, maintaining WRI > 0.70. *Paraphrasing* (λ=0.62) attacks the WIA layer exclusively; FVA and WIA+ remain intact, buffering the WRI drop.

Cross-layer interaction and WRI design. The approximate additive behavior of WRI ($p = 0.62$ for full vs. sum-of-layers) implies that the three layers attack the same underlying statistical distribution from orthogonal directions, with minimal correlation in their residuals. This orthogonality is by design: WIA operates on the output probability simplex, FVA on the hidden-state cosine space of trigger responses, and WIA+ on the weight-parameter ℓ^2 ball. An adversary must simultaneously attack all three subspaces to defeat GuardMark, whereas defeating any single layer costs only one attack budget level λ while the remaining layers compensate. The $\gamma_w = 0.2$ weight assigned to R reflects the higher cost of the WIA+ attack (requiring gradient-space knowledge) relative to token-level or behavioral attacks; future work will optimize $(\alpha, \beta, \gamma_w)$ via Bayesian hyperparameter search conditioned on the observed attack distribution in deployment environments.

Mobile and edge deployment. GuardMark's 3.2%–4.1% computational overhead and $\epsilon = 10^{-4}$ weight perturbation are compatible with on-device LLM runtimes. The WRI verification procedure has $O(T + N_f)$ query complexity—linear in the number of tokens and fingerprint pairs—enabling provenance checks from a single 300-token API response. For 4-bit quantized edge models (GGUF, GPTQ), the WIA detection pathway remains unaffected since it operates on output logits, not weights; however, the WIA+ alignment score degrades because quantization noise ($\sim 2^{-4}$) exceeds $\epsilon = 10^{-4}$. Increasing ϵ to 10^{-3} would restore WIA+ robustness at the cost of a slight imperceptibility trade-off, which we leave to future work.

VIII. Conclusion

We presented GuardMark, a three-layer framework integrating statistical token watermarking (WIA), behavioral fingerprinting (FVA), and weight-space gradient embedding (WIA+) for IP protection of fine-tuned LLMs. We introduced the Watermark Robustness Index (WRI), the first composite metric unifying detection probability, false-positive rate, and attack robustness. Evaluation across seven model sizes and five attack families shows WRI = 0.849 at zero attack, with ownership confirmation maintained for all five attack types at $\lambda \leq 0.50$ and 3.2% computational overhead— outperforming all baselines on every measured dimension.

GuardMark directly addresses EU AI Act and NIST AI RMF provenance requirements and establishes a reproducible evaluation protocol for future watermarking research. The WRI score provides a single decision-ready statistic compatible with forensic workflows: because each layer probes a structurally orthogonal subspace (token distribution, hidden-state cosine space, and weight-parameter ℓ^2-ball), a court or auditor can inspect individual layer scores to understand *why* ownership was confirmed or denied, rather than treating the decision as a black box. This transparency property distinguishes GuardMark from prior single-layer approaches and is essential for legal admissibility in IP disputes.

A. Limitations and Future Work

Simulation scope. The main evaluation is simulation-based, calibrated on a single real-model run (GPT-2, 125 M). Simulation error bounds (0.7% on $\hat{\gamma}$, 3.1% on $\bar{z}$) are validated only at 125 M; scaling behavior at 7 B and 13 B parameters requires further empirical validation. Future work will deploy GuardMark on full LLaMA-2 and Mistral checkpoints.

4-bit quantization and open-weight models. The current evaluation covers 8-bit quantization. 4-bit quantization (GGUF, GPTQ)—prevalent in edge and mobile LLM deployment—may more aggressively degrade the WIA+ weight-space signature and will be evaluated in follow-up work. For open-weight base models, adversaries can compute the weight difference $\mathcal{M}^* - \mathcal{M}$ directly; future work will explore key-oblivious embedding strategies to close this gap.

Ownership threshold. The $\tau_{\mathrm{WRI}} = 0.70$ threshold was derived as the ROC-optimal point on the current simulation set. Legal admissibility of WRI scores as IP ownership evidence will depend on jurisdiction-specific standards not yet codified under the EU AI Act implementation acts. Calibrating τ_{WRI} on larger, heterogeneous attack suites and validating with third-party red-teaming remain important directions before WRI-based claims are admitted as forensic evidence in court proceedings.

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